A Report on Behavioural Research in Climate Policy, based on Andre et al. 2024

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1 Introduction

Given the threats of global warming, as consolidated in the IPCC (2022) report, it is essential to understand the behavioural drivers of climate change and design effective environmental policies and international agreements to tackle it. Therefore, policymakers deciding on those environmental policies should have access to all the necessary information about human behaviour, including beliefs, attitudes, motivations and social interaction, to design them in a way that the general public and institutions accept.

In this context, it is essential to understand what determines an individual's willingness to fight global warming. Using Andre et al. (2024b) as a starting point, this report analyses the role of perceived social norms in influencing climate behaviour and the potential benefits of correcting misperceptions in terms of improving citizens' support of current and upcoming EU climate regulations.

2 Summary of 'Misperceived Social Norms and Willingness to Act Against Climate Change' by Andre et al.

2.1 Research Question and Design

Andre et al. (2024b) addressed this issue by examining how individuals perceive social norms and how this perception affects their willingness to fight global warming. The study used a two-wave design to identify a causal effect of norm perception on climate behaviour.

In Wave 1 (N=2,000, March 2021), Andre et al. (2024b) collected baseline data on perceived social norms through a stratified survey targeting the US adult population. Respondents answered key questions about their own climate behaviour and how they perceived the climate behaviour of others. These responses act as the actual prevalence rates against which misperceptions could later be measured. Respondents also completed incentivised donation decisions, dividing \$450[1] between themselves and *atmosfair*, a climate action charity. Additionally, the authors collected detailed information on economic preferences, moral foundations and baseline demographic characteristics.

In Wave 2 (N=6,000, April 2021), a randomised controlled trial tested whether correcting misperceptions causally influences climate behaviour. Respondents were randomly assigned to one of three groups: a Control group, a Behaviour treatment group, who were told that 62% of Americans try to fight global warming and a Norms treatment group, who were told that 79% of Americans believe people in the US should fight global warming (actual shares from Wave 1). All respondents then made an incentivised donation decision identical to Wave 1. After the treatment, they also reported support for six specific climate-friendly behaviours and support for climate policies on a four-point Likert scale.

2.2 Main Results and Conclusions

Perceived social norms are statistically significant predictors of climate behaviour, with effect sizes comparable to moral values and economic preferences such as altruism and reciprocity. It was found that Americans systematically underestimate the prevalence of climate-friendly norms. The information experiment then showed that informing respondents about the actual prevalence reduced these misperceptions and encouraged individuals to adopt climate-friendly behaviour.

[1] Approximately the annual CO_2 emissions of a typical US citizen.

This effect is especially large among participants who are sceptical about global warming, a group that is generally difficult to reach.

To Sum Up

Many people hold pro-climate views but mistakenly believe others do not. Correcting this misperception can trigger a positive feedback loop, so learning about actual majority support encourages visible climate action, which in turn reinforces social norms and encourages others to follow suit.

Andre et al. (2024b) summarised their main findings in an early policy letter (Andre et al. 2021), which was aimed at decision-makers in politics and government. In addition, a concurrent study by Andre et al. (2024a) provided complementary cross-country evidence for these results in a worldwide context, making information treatments a potentially scalable and low-cost complement for prompting and scaling climate-friendly behaviour at the EU or worldwide level.

3 Positioning within Course Themes and Literature

3.1 Central Behavioural Mechanism: Perceived Social Norms

The central behavioural mechanism studied by Andre et al. (2024b) is a causal chain. Individuals hold systematically misperceived beliefs about how many Americans actually try to fight global warming (behaviour) and how many think one should do so (norms). Correcting either belief corrects their perceived social norm, raising their own support for climate action and policies.

Ganguli and Mengel (2026) demonstrate that climate decisions are interdependent rather than autonomous, so individuals' willingness to reduce emissions responds to social influence, making beliefs about others a key input into their own decision-making process. Consequently, cooperation is limited by the accuracy of the beliefs upon which it relies. This 'pluralistic ignorance'[2] ties climate action to a misperceived pessimistic reference point.

Importantly, the central behavioural mechanism (stated by Andre et al. 2024b) must also be distinguished from conventional behavioural interventions, e.g. nudges. Gravert and Olsson Collentine (2021) contrast simple nudges with normative and monetary treatments in a public-transport field experiment and find that nudges alone rarely generate lasting behavioural shifts. They state that a nudge alters the choice architecture surrounding a decision, but leaves the underlying reasons for the behaviour untouched, so its effect decays once the intervention is withdrawn. In contrast to nudges Andre et al. (2024b) target this underlying layer. By correcting misperceived social norms, they intervene in an individual's fundamental behaviour rather than the framing of a choice. In principle, this makes it self-sustaining, since a corrected belief about others does not need to be repeatedly re-administered. Whether this effect is self-sustaining remains an open question, since treatment effects are typically measured immediately after information is provided. Media and polarised debates can quickly distort perceptions again.

Crucially, this central behavioural mechanism also operates in a political domain, where correcting norm perceptions also raises stated support for climate policies (Andre et al. 2024b). This raises the question whether corrected norm perceptions can also translate into support for binding policy instruments.

[2] Pluralistic ignorance describes a situation in which individuals believe their private views to deviate from those of the majority, leading them to suppress those views in order to conform to a perceived social norm (Pilat and Krastev 2023).

3.2 Relation to Behavioural Climate Policy Literature

The focal paper addresses the political economy of climate policy, where the low public support for efficient instruments like carbon pricing remains a central puzzle. Traditional policy design assumes that voters are driven primarily by monetary payoffs. However, Mildenberger, Lachapelle, et al. (2022) demonstrated that rebates, such as returning carbon-tax revenue, fail to reliably improve public support[3]. This highlights that monetary compensation alone is an insufficient instrument for increasing support for climate policy. Existing literature typically views behavioural tools as downstream complements to fix friction under existing price signals (Gravert and Shreedhar 2022)[4]. But correcting misperceived social norms targets the political acceptability of such policy instruments in the first place, raising stated support for climate policies.

In summary, the primary contribution of the paper and the policy letter (Andre et al. 2021) is causally demonstrating that correcting misperceptions about social norms raises willingness to act and policy support, especially among climate sceptics. It bridges the gap between behavioural norm theory and scalable, cost-effective climate policy interventions. However, correcting misperceptions does not eliminate the core global public good problem, where free-riding incentives persist across nations. Whether behavioural interventions can build public backing for the binding, internationally coordinated mechanisms required to solve this structural failure is a matter of policy design, to which the following section turns.

Key Message

Conditional cooperation is only as accurate as the beliefs it conditions on. The binding constraint on climate action is therefore not the absence of climate-friendly norms, but their systematic misperception.

4 Policy Relevance and Regulatory Context

4.1 Policy Domain and Regulatory Landscape

The main results of Andre et al. (2024b) established that individual willingness to act is deeply embedded in social expectations, but constrained by widespread pluralistic ignorance. This misperception is not unique to the US, as global replication evidence (Andre et al. 2024a) suggests the pattern also extends to the EU. To evaluate the practical value of this insight for EU policymakers, however, it must be situated within the current regulatory landscape of EU climate policy.

Historically, instruments such as the EU Emissions Trading System (ETS) targeted large point-source emitters, keeping the immediate costs of decarbonisation largely hidden from consumers. The European Green Deal instruments now directly affect household budgets and behaviour. The revised Energy Performance of Buildings Directive[5] (EPBD) obliges Member States to decarbonise their building stock. In addition, the second Emissions Trading System[6] (ETS2) introduces explicit carbon pricing for heating and transport. Both instruments are already legally adopted and under active implementation, moving climate policy from an abstract, industry-level issue to a visible, consumer-facing one where public acceptance becomes a binding constraint.

[3] Main finding from Sophia Kaindl's presentation, 15 June 2026.
[4] Discussed in Jonas Lehmköster's presentation, 15 June 2026.
[5] The revised EPBD aims at a fully decarbonised building stock by 2050 and thus contributes directly to the EU's energy and climate targets.
[6] ETS2 covers CO_2 emissions from fuel combustion in buildings, road transport, and additional sectors.

This shift makes the domain highly relevant to Andre et al. (2024b). If citizens systematically underestimate climate-friendly behaviour and approval, they incorrectly assume a public consensus against climate action. This misperception artificially dampens public support for precisely the kind of robust pricing policies and international climate clubs that Ockenfels (2023) advocates. Worse, it creates space for vocal minorities to dominate the public narrative and for policymakers to underestimate feasible ambition. As recently observed in European debates over green heating requirements, a narrative emerged that was amplified by the disproportionate media presence of a vocal minority. This narrative suggested that there was public opposition to green heating requirements. This suppressed norm enforcement and discouraged private investment in technologies such as heat pumps.

Crucially, this constraint does not only affect citizens. Mildenberger and Tingley (2019) document that policymakers themselves systematically misperceive public support for climate action. By mistaking vocal resistance for the median voter, politicians anticipate electoral punishment. Misperception constrains both citizens and policymakers, producing an equilibrium in which majority support fails to translate into policy. This dynamic stalls legislative implementation, reduces ambition and increases political polarisation.

4.2 Refining the Policy Mix

While norm correction builds domestic political mandates, international climate cooperation fails structurally. National sovereignty limits enforcement and voluntary pledges predictably fail to overcome free-riding (Ockenfels 2023; Cramton et al. 2017).

Despite this evidence, the systematic use of norm-based insights at the regulatory level is mostly absent. Institutional communication within the European Union, such as the European Climate Pact, still dominantly emphasises scientific urgency and the physical consequences of inaction. This represents a missed opportunity, since existing literature tends to frame behavioural tools merely as downstream complements that fix friction under existing price signals (Gravert and Shreedhar 2022). By correcting the narrative about what fellow citizens actually do and endorse, policymakers could instead directly target the political acceptability of policies like ETS2 and EPBD.

Applying the insights of Andre et al. (2024b) to this case provides a refinement that complements financial incentives. Rather than substituting economic instruments, behavioural insights should establish their political preconditions. Policymakers can eliminate a significant cause of perceived financial penalty that fuels public resistance and legislative backlash. Thus, norm correction could create the political space in which carbon pricing and regulatory mandates gain legitimacy and durability.

Therefore, norm correction should be understood as a complement to, rather than a substitute for, conventional instruments such as the ETS2 and EPBD. This is consistent with Gravert and Shreedhar (2022), who argues that behavioural tools and carbon pricing reinforce one another. Instead of solely communicating the technical necessities or financial subsidies of new regulations, these campaigns should systematically highlight existing social support. For instance, messaging such as '79% of EU citizens believe we should fight climate change' or '> 50% of homeowners in your region support the transition to renewable heating' transforms compliance from an individual financial burden into a shared collective endeavour. While this does not eliminate the material costs of decarbonisation, it reframes them, stabilising long-term public support, insulating regulations from short-term political backlash, and ultimately increasing the political feasibility of ambitious pricing regimes.

4.3 Limitations and Avenues for Future Research

While the experimental evidence by Andre et al. (2024b) is robust, applying these insights to the real-world policy domains of ETS2 and EPBD reveals limitations.

First, correcting misperceptions cannot overcome structural or financial barriers. The dependent variable in the authors' experiment involves small-scale donations. In contrast, complying with the EPBD requires households to invest over $10,000 in heat pumps or insulation. Even if an individual correctly perceives a strong climate-friendly norm, liquidity constraints and split-incentive problems (e.g., landlord-tenant dilemma) will prevent action if not accompanied by subsidies such as the EU Social Climate Fund.

Also, the focal paper isolates one specific behavioural market failure. Climate inaction is driven by multiple factors, including present bias or limited attention. Correcting normative misperceptions addresses none of these directly.

Third, because outcomes are elicited immediately after the information provision, the experiment establishes the existence of a belief-based channel. It remains unclear whether a single informational message can durably shift perceived norms, or whether the effect dissipates once individuals return to a real-world media ecosystem characterised by polarised narratives. This is the environment that produced the misperception in the first place. In such polarised landscapes, information campaigns by state authorities might also trigger reactance among certain voter segments rather than correction.

Scaling these insights into actionable policy requires moving from online surveys to field experiments, such as evaluating how normative information affects behaviour when individuals face real rebates for heat pumps under the EPBD or actual carbon costs under ETS2. But this presupposes an infrastructure that does not yet exist. Because norm correction can only communicate what has previously been measured, it requires close collaboration among the institutes, government, and NGOs to collect representative benchmark data and to evaluate campaigns in the field. A second promising effect is the heterogeneity of the treatment effect. Since the focal paper finds the strongest responses among sceptics and among those with low intrinsic willingness to act, future studies should test tailored communication strategies for specific demographic and political groups, and examine how such interventions can be systematically integrated into national climate action plans.

Key Message

Correcting misperceived norms does not replace carbon pricing—it makes it politically survivable.

5 Conclusion

This report has examined the pivotal role of perceived social norms in climate policy, based on the empirical findings of Andre et al. (2024b). The paper's contribution to behavioural climate research is that it identifies 'pluralistic ignorance', the systematic underestimation of fellow citizens' willingness to act and establishes causally that correcting this misperception raises both climate-friendly behaviour and stated policy support. Positioning the paper within the broader literature has shown that individual climate behaviour is neither purely a response to economic incentives nor purely a matter of global macro-cooperation, but is deeply embedded in expectations and conditional cooperation.

The practical relevance of these findings is defined by the current regulatory context in the

European Union. As instruments like the revised EPBD and the ETS2 begin to impose visible costs on households, the binding constraint shifts from instrument design to public acceptance. Norm correction provides the empirical support needed to dissolve the phantom majority of opponents and secures the 'democratic licence to regulate'. Because norm correction is low-cost and scalable, policymakers can integrate such campaigns into EU-wide, national and communal communication strategies without substantially increasing administrative burden.

But the approach has clear limits. Correcting misperceived norms cannot overcome structural barriers such as liquidity constraints or the split-incentive problems inherent in landlord-tenant relationships. Nor can it address other behavioural market failures that independently impede climate action. Most importantly, the experimental evidence establishes the existence of a belief-based channel, but it remains unclear whether a single informational message can sustain shifted norms once individuals return to polarised real-world media ecosystems that originally produced the misperception. Overcoming those limitations requires field experiments and evaluating how normative information affects behaviour when households face actual carbon costs or real renovation rebates, not hypothetical small donations.

The behavioural mechanism that emerges as particularly important for policy design is **debiasing through transparent norm communication**. Its value lies not in the delivery channel, but in the level at which it operates. By correcting the beliefs on which conditional cooperation rests, norm correction addresses a foundational constraint on collective action. In this sense, behavioural policy is less a corrective to individual irrationality than a precondition for coordinated transition. It does not reduce the cost of decarbonisation, nor does it resolve the international free-riding problem that binding coordination must address. What it offers is a scalable, cost-effective tool to align individual perceptions with existing majority support. This creates the political space in which structural economic transitions can gain legitimacy and durability.

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